

Q1) Describe the system chosen in free text. Clarify the input and output domains. [20%]

Answer:

The chosen system is a loan assessment based on credit history, age, and salary level.

The output is the Loan Approval Score, indicating the probable loan approval.

Inputs Domain:

SN	Input Name	Range	Reason
1	Credit Score	300–850	Real-life credit score ranges from 300 to 850
2	Income	0–120,000\$/year	Annual gross earnings from minimum wage to upper-middle class, split into Low, Medium, and High.
3	Age	8–70 years	Working-age adults only, split into Young (25), Middle-aged (40), and Old (~60).

Output:

SN	Output Name	Range	Reason
1	Loan Approval Score	0-100	Indicating whether the loan should be approved or not

Define the input and output variables, providing for each linguistic variable the fuzzy values and the graphical representation of their membership functions. Motivate their interval(s). Choose AND function as the min operator; aggregation and OR as the max operator, and Gaussian/sigmoidal membership functions. You are also expected to submit the fuzzy inference system (FIS) file of the system you designed. [20%]

Linguistic Variable, Credit Score			
Inputs	Notation	Numerical Range	Numerical Range(Normalized)
Low	L	300 – 500	0.0 – 0.4
Medium	M	450 – 700	0.3 – 0.7
High	H	650 – 850	0.6 – 1.0

Linguistic Variable, Age			
Inputs	Notation	Numerical Range	Numerical Range(Normalized)
Low	L	18 – 30	0.0 – 0.3
Medium	M	25 – 50	0.2 – 0.7
High	H	45 – 65	0.6 – 1.0

Linguistic Variable, Income			
Inputs	Notation	Numerical Range	Numerical Range(Normalized)
Low	L	0 – 30,000	0.0 – 0.3
Medium	M	25,000 – 70,000	0.2 – 0.7
High	H	60,000 – 120,000	0.6 – 1.0

Linguistic Variable, Eligible for Loan			
Inputs	Notation	Numerical Range	Numerical Range(Normalized)
Low	L	0 – 40	0.0 – 0.4
Medium	M	30 – 70	0.3 – 0.7
High	H	60 – 100	0.6 – 1.0

```

% Create FIS
fis = newfis('loan_approval', 'mamdani');

% add new variable
fis = addvar(fis, 'input', 'CreditScore', [300,850])
a = 850-300
interval = a/3
val = 350
fis = addmf(fis,'input', 1, 'Low', 'gaussmf', [40,val])
fis = addmf(fis,'input', 1, 'Medium', 'gaussmf', [40,val + interval])
fis = addmf(fis,'input', 1, 'High', 'gaussmf', [40,val + 2*interval])
plotmf(fis, 'input', 1);
waitforbuttonpress;

fis= addvar(fis, 'input', 'Income', [0,120000]);
fis=addmf(fis,'input', 2, 'Low', 'gaussmf', [10000,20000]);
fis=addmf(fis,'input', 2, 'Medium', 'gaussmf', [10000,60000]);
fis=addmf(fis,'input', 2, 'High', 'gaussmf', [10000,100000]);
plotmf(fis, 'input', 2);
waitforbuttonpress;

fis = addvar(fis, 'input', 'Age', [18,70]);
fis = addmf(fis,'input', 3, 'Young', 'gaussmf', [3,25]);
fis = addmf(fis,'input', 3, 'Middle-aged', 'gaussmf', [5,40]);
fis = addmf(fis,'input', 3, 'Old', 'gaussmf', [4,60]);
plotmf(fis, 'input', 3);
waitforbuttonpress;

writefis(fis, 'Q2_loan_approval_fis');

fis = addvar(fis, 'output', 'LoanApproval', [0 100]);
fis = addmf(fis, 'output', 1, 'Low', 'gaussmf', [15 25]);
fis = addmf(fis, 'output', 1, 'Medium', 'gaussmf', [15 50]);
fis = addmf(fis, 'output', 1, 'High', 'gaussmf', [15 75]);

```

Figure: Octave code to add membership function to a inputs and plotting it

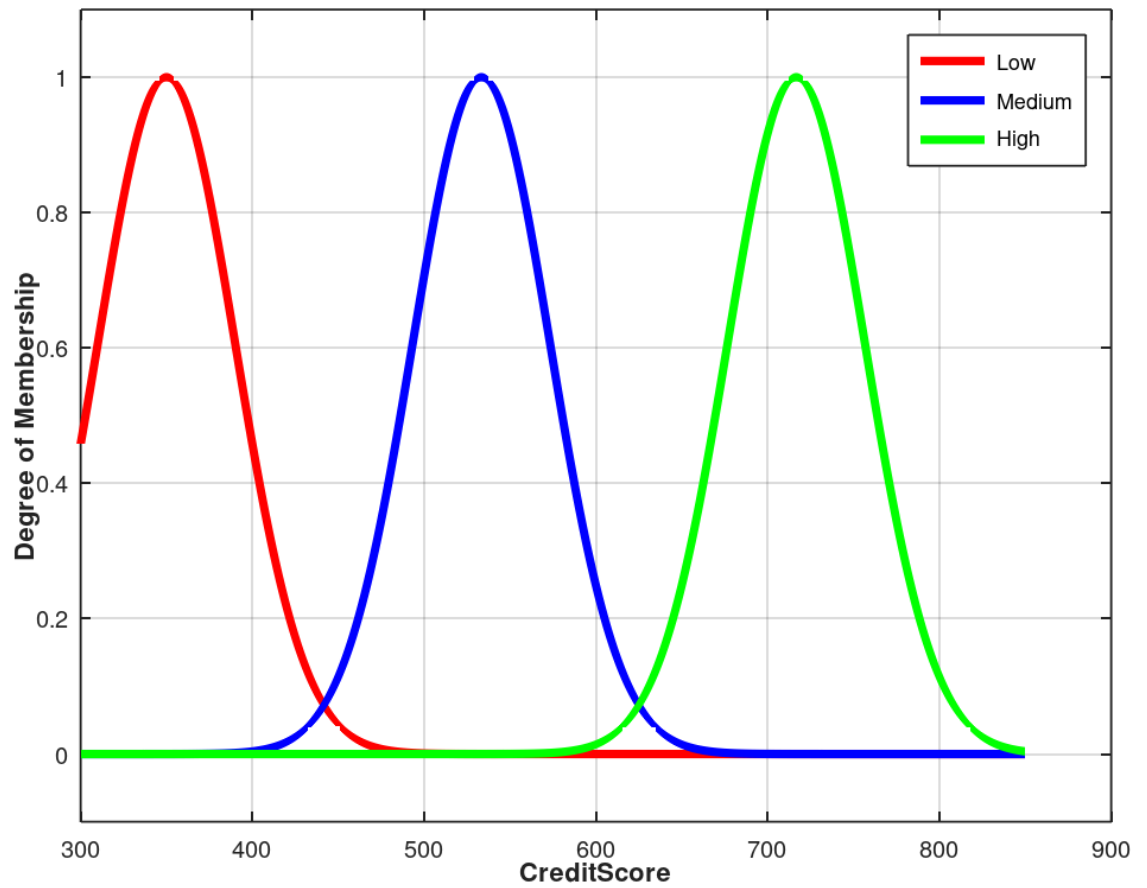


Figure: Membership function for credit score

Similarly for the other two inputs:

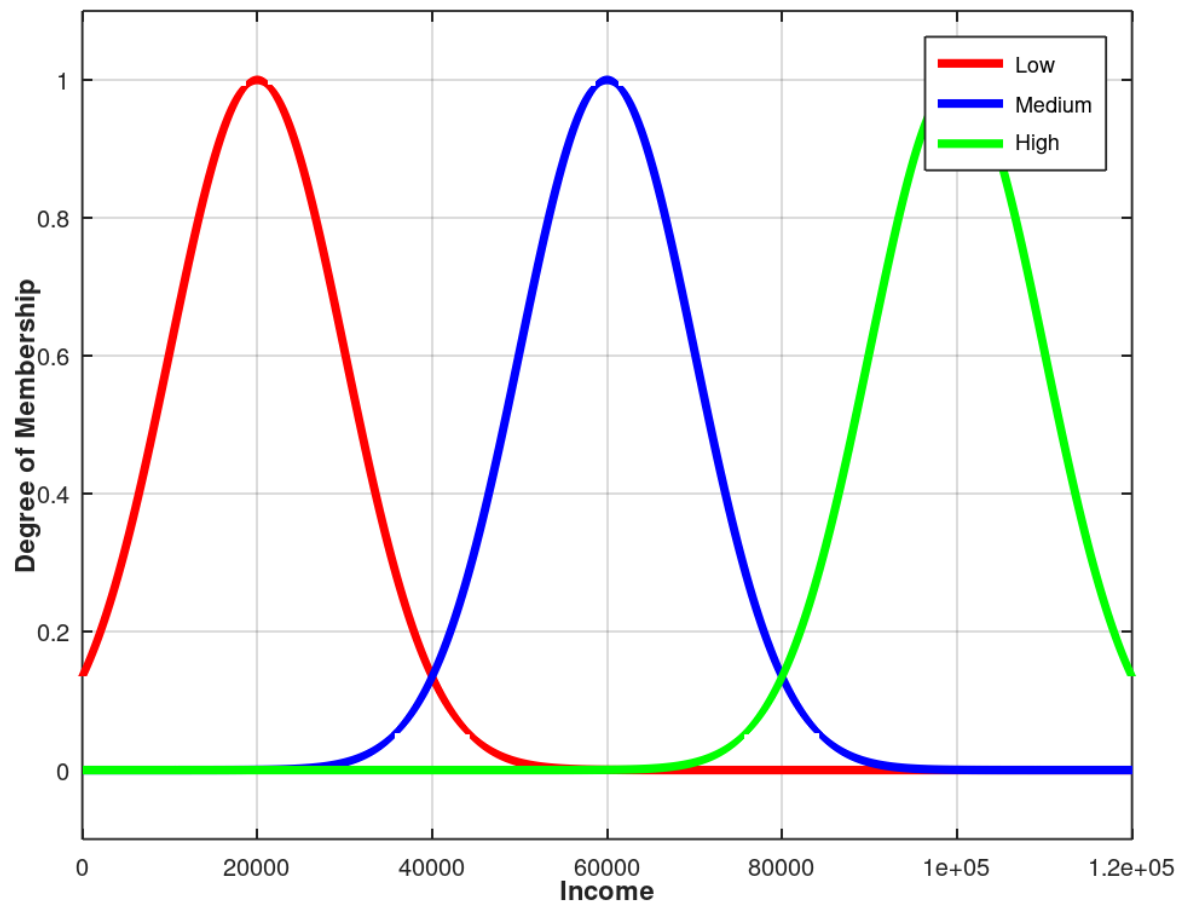


Figure: Membership function for Income

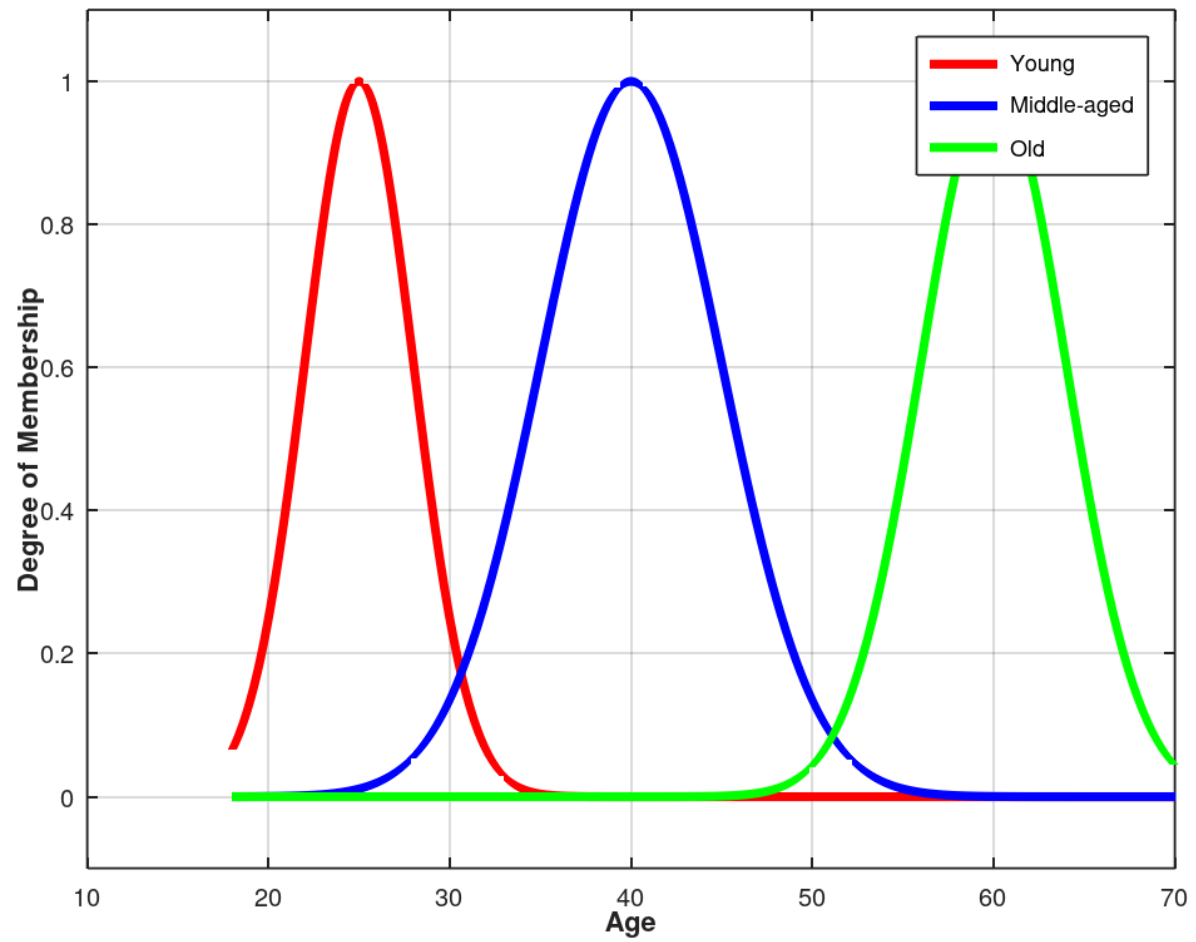


Figure: Membership function for Age

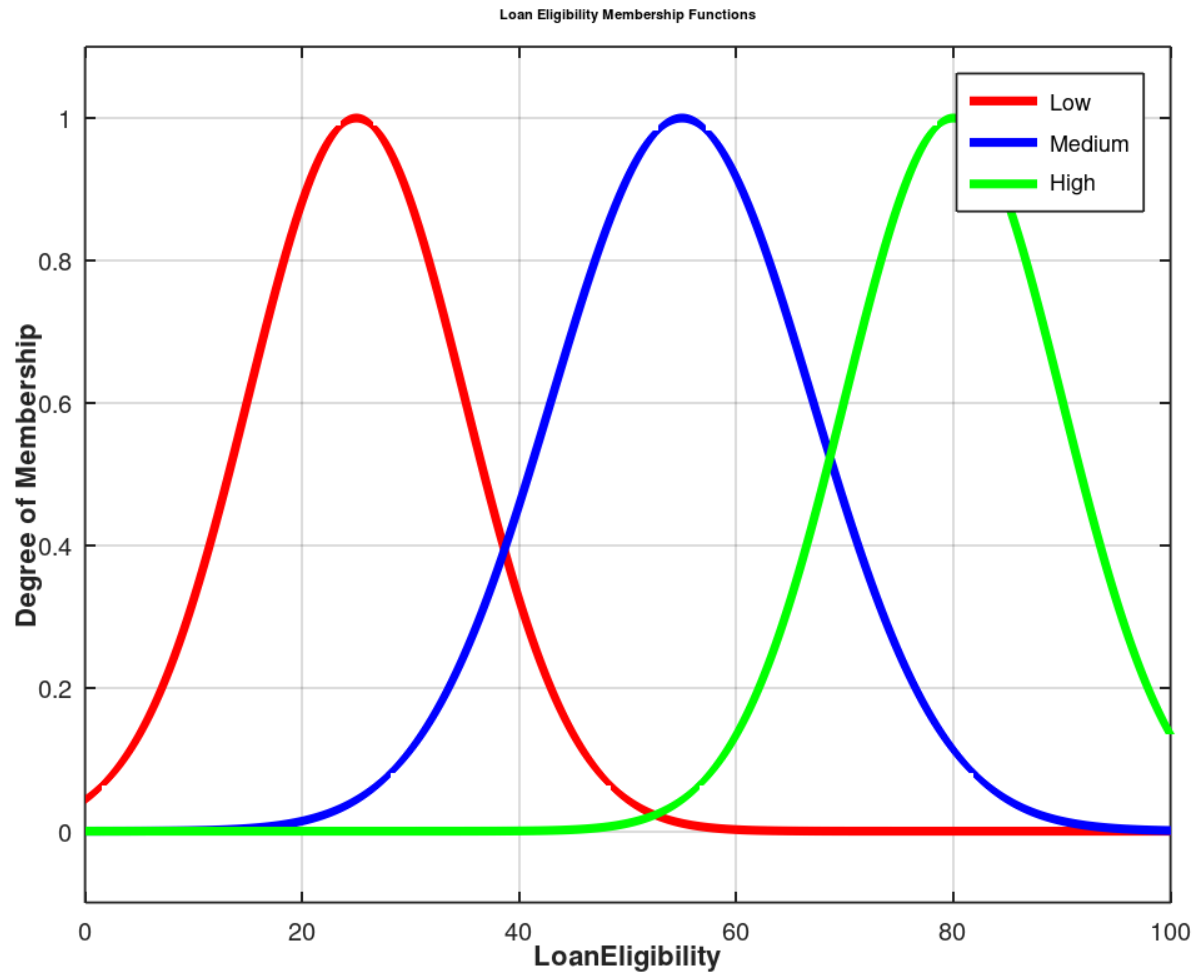


Figure: Loan Eligibility Membership Function for Output

Addition of rule:

Rules:

1. If CreditScore is High AND Age is Old, then LoanEligibility is High
2. If CreditScore is Medium AND Age is Middle, then LoanEligibility is Medium
3. If CreditScore is Low, then LoanEligibility is Low
4. If CreditScore is High, then LoanEligibility is High
5. If CreditScore is High, then LoanEligibility is High
6. If Age is Young, then LoanEligibility is Low
7. If CreditScore is Medium AND Income is Medium AND Age is Middle, then LoanEligibility is Medium
8. If CreditScore is Low AND Income is Medium AND Age is Middle, then LoanEligibility is Low

In octave, we do it by:

```
octave:48> ruleList = [  
3 3 3 3 1 1; % High credit, high age, high income → High eligibility  
3 3 2 3 1 1;  
3 2 3 3 1 1;  
3 2 2 2 1 1;  
  
2 3 3 3 1 1;  
2 2 3 2 1 1;  
2 2 2 2 1 1;  
2 1 2 2 1 1;  
  
1 1 1 1 1 1;  
1 2 1 1 1 1;  
1 1 2 1 1 1;  
1 2 2 2 1 1;  
  
3 1 3 3 1 1;  
1 3 3 2 1 1;  
];  
  
fis = addrule(fis, ruleList);  
octave:50> █
```

Figure: Addition of rules in Octave

These rules can be visualized by showrule command in octave.

```
octave:54> showrule(fis)
1. If (CreditScore is High) and (Age is High) and (Income is High), then (LoanEligibility is High) (1)
2. If (CreditScore is High) and (Age is High) and (Income is Medium), then (LoanEligibility is High) (1)
3. If (CreditScore is High) and (Age is Medium) and (Income is High), then (LoanEligibility is High) (1)
4. If (CreditScore is High) and (Age is Medium) and (Income is Medium), then (LoanEligibility is Medium) (1)
5. If (CreditScore is Medium) and (Age is High) and (Income is High), then (LoanEligibility is High) (1)
6. If (CreditScore is Medium) and (Age is Medium) and (Income is High), then (LoanEligibility is Medium) (1)
7. If (CreditScore is Medium) and (Age is Medium) and (Income is Medium), then (LoanEligibility is Medium) (1)
8. If (CreditScore is Medium) and (Age is Low) and (Income is Medium), then (LoanEligibility is Medium) (1)
9. If (CreditScore is Low) and (Age is Low) and (Income is Low), then (LoanEligibility is Low) (1)
10. If (CreditScore is Low) and (Age is Medium) and (Income is Low), then (LoanEligibility is Low) (1)
11. If (CreditScore is Low) and (Age is Low) and (Income is Medium), then (LoanEligibility is Low) (1)
12. If (CreditScore is Low) and (Age is Medium) and (Income is Medium), then (LoanEligibility is Medium) (1)
13. If (CreditScore is High) and (Age is Low) and (Income is High), then (LoanEligibility is High) (1)
14. If (CreditScore is Low) and (Age is High) and (Income is High), then (LoanEligibility is Medium) (1)
```

Figure: Visualization of 'showrule' command in octave

Q3) Present an inference for output Y graphically, where X1, X2 and X3 have given values of your choice
("graphically" means using Rule Viewer and setting the values in it). Interpret the results. [20%]

Inference for the output:

```
output = evalfis([400 50000 30], fis);
printf('Loan Approval: %.2f\n', output);
```

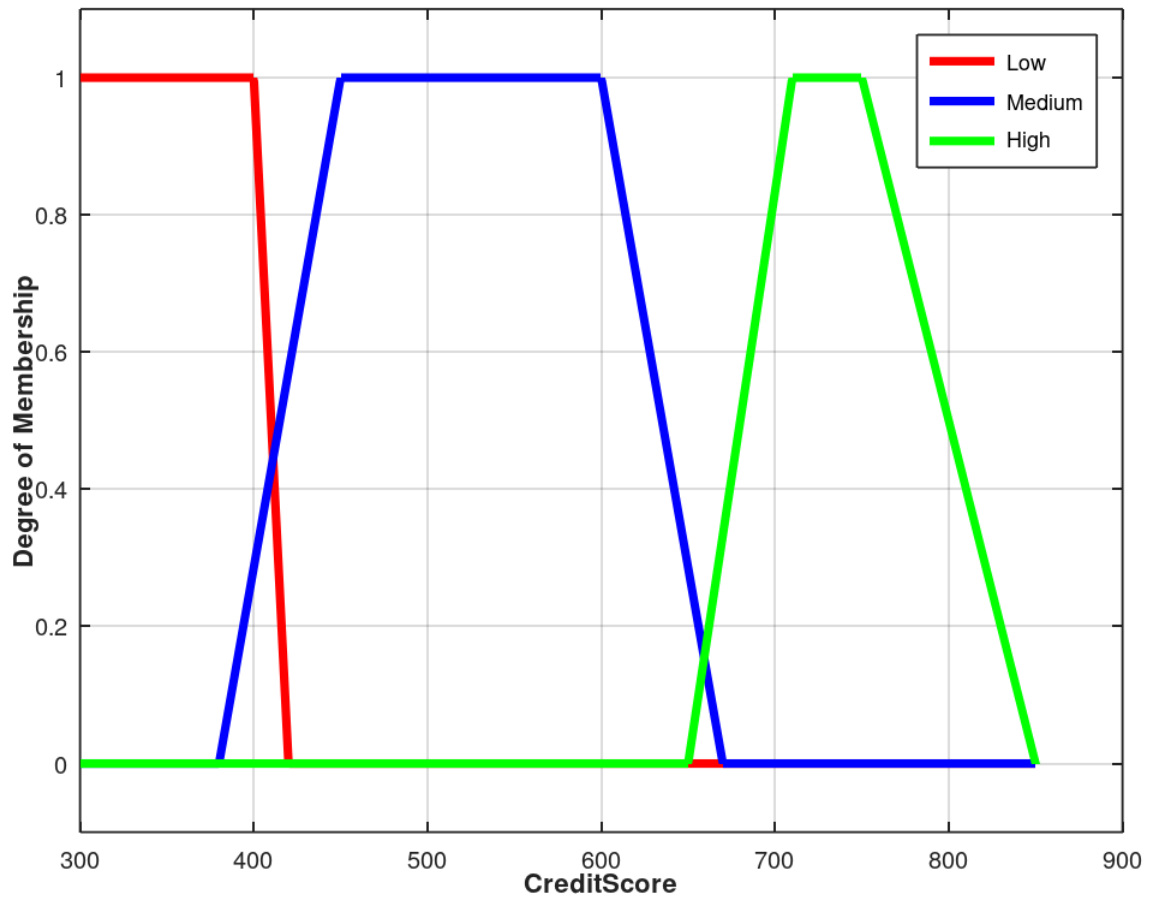
The output is:

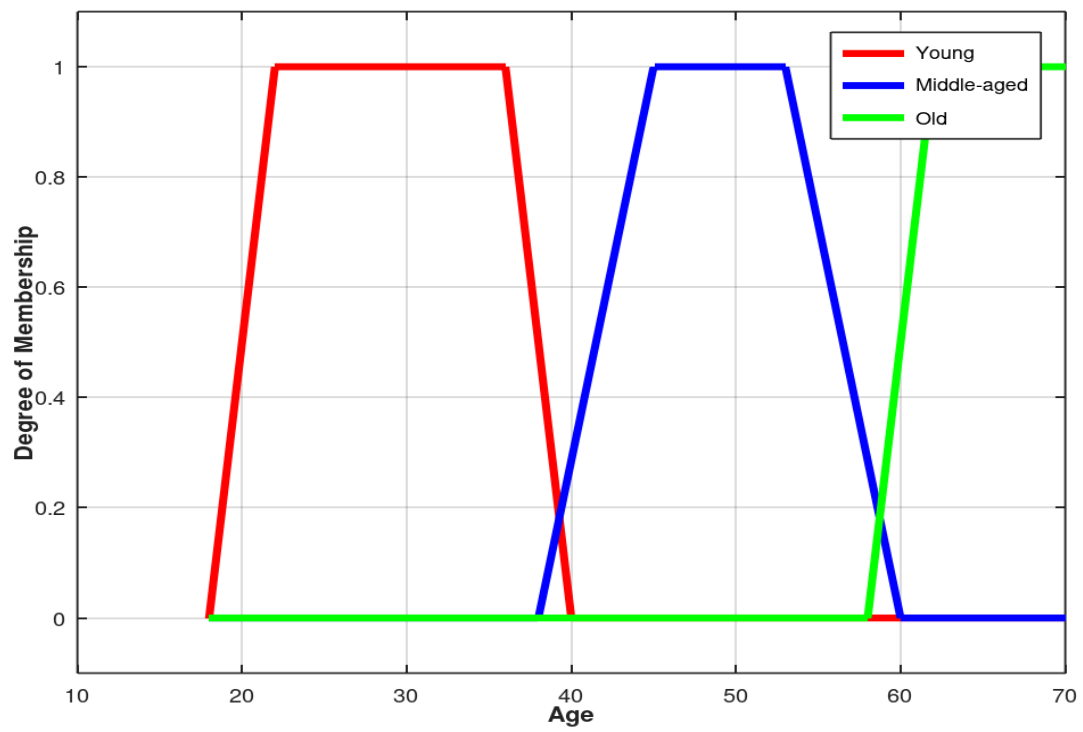
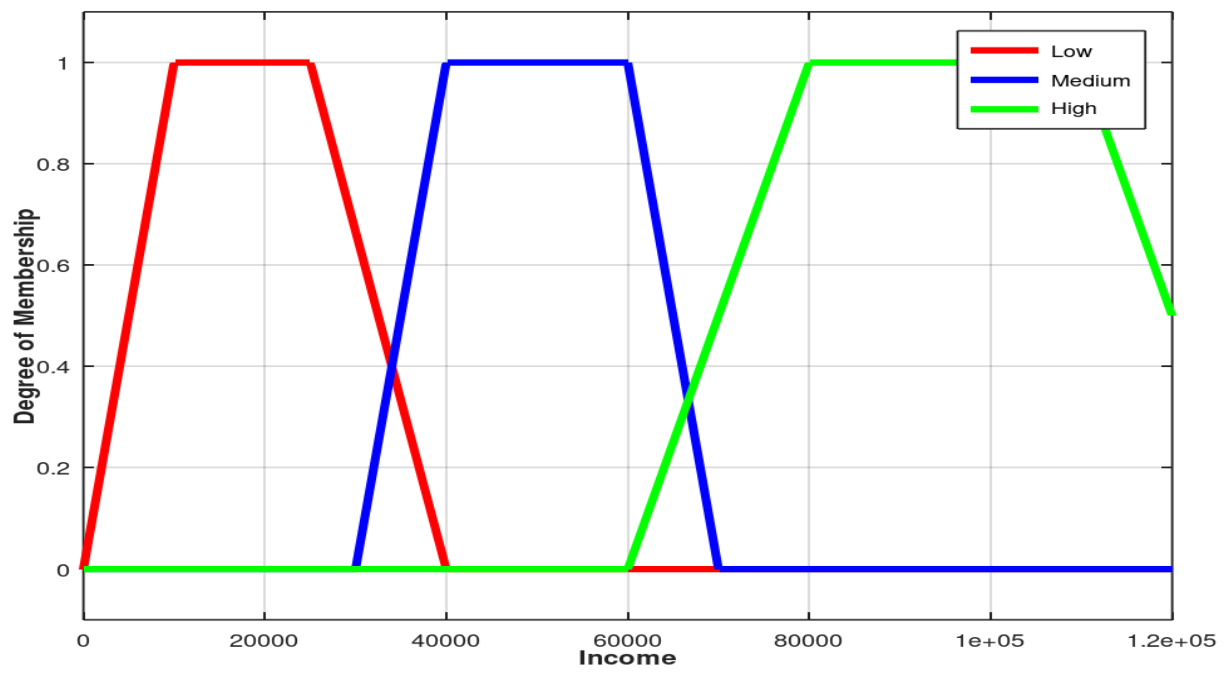
```
Loan Approval: 29.03
```

This suggests that a person with a 400 credit score, with 50000 income, and age 30 is 29% likely to get the loan approval. The result occurred because of his low income range.

4) Draw the output surface and briefly describe and discuss changes relative to the original system output surface for the case when the Trapezoidal membership function (trapmf) is used as a membership function for all system variables. [20%]

Changes in surfaces:





3 High) (1)
Loan Approval: 31.26
Method: centroid

The loan approval shown is 31.26, slightly higher than the Gaussian, as it has a larger straight line covering area.

5) Investigate the different defuzzification techniques available in Octave Fuzzy Logic Toolbox and carry out an analysis of their effects on the output. [20%].

Analysis

In the Gaussian membership function:

```
Method: centroid      Result: 56.98
Method: bisector      Result: 58.00
Method: mom           Result: 60.00
Method: som           Result: 20.00
Method: lom           Result: 100.00
```

Other defuzzification methods yield more accurate results, such as the centroid method.

SN	Method	Result	Analysis
1	Centroid	50	This gives a balanced value because it considers the entire shape of the fuzzy set. It feels like the “fair middle point” of the result.
2	Bisector	51	Very close to centroid, but slightly shifted. This happens because it splits the area into two equal halves rather than balancing the whole shape.
3	mom	50	Since it only looks at the highest membership values, it lands right around the peak. Works well when the top is flat or symmetric.
4	som	35	This picks the smallest value among the maximum membership region, so it leans toward the lower side of the

			output.
5	lom	65	Opposite of SOM — it picks the highest value in the max region, so the result shifts toward the upper side

In the Trapezoidal functions:

```
Method: centroid      Result: 50.00
Method: bisector      Result: 51.00
Method: mom           Result: 50.00
Method: som           Result: 35.00
Method: lom           Result: 65.00
```

SN	Method	Result	Analysis
1	Centroid	50	Still gives a balanced result. Even though the shape is different, it still finds the “center of mass” of the output.
2	Bisector	51	Slightly different from the centroid but still very close. This shows the output distribution is fairly even
3	mom	50	The flat top of the trapezoid makes MOM stable, so it ends up similar to the centroid here.
4	som	35	This pulls the result toward the left edge of the highest region, giving a more conservative output.
5	lom	65	This pushes the result to the right edge, making it more optimistic or higher-valued.